A Cyber-War Between Bots: Cognitive Attackers are More Challenging for Defenders than Strategic Attackers

YINUO DU, Software and Social Systems Department, Carnegie Mellon University, Pittsburgh, United States
BAPTISTE PREBOT, Department of Social and Decision Sciences, Carnegie Mellon University, Pittsburgh, United States
TYLER MALLOY, Department of Social and Decision Sciences, Carnegie Mellon University, Pittsburgh, United States
CLEOTILDE GONZALEZ, Department of Social and Decision Sciences, Carnegie Mellon University, Pittsburgh, United States

Adversary emulation is commonly used to test cyber-defense performance against known threats to organizations. However, many adversary emulation methods often rely on automated planning and underplay the role of human cognition. Consequently, defenders are often underprepared for human attackers who can think creatively and adapt their strategies. In this article, we propose the design of adversarial cognitive agents that are dynamic, adaptable, and able to learn from experience. These cognitive agents are built based on the theoretical principles of Instance-Based Learning Theory (IBLT) of experiential choice in dynamic tasks, making them more challenging than strategically optimal adversaries for human defenders. Our research offers three main contributions. First, in a simulation experiment, we demonstrate how IBL attacker agents can learn from experience and become as efficient as optimal strategic algorithms against a strategic defender. In a second simulation experiment, the IBL attackers are pitted against an IBL defender, showing that the IBL attacker can be a more challenging adversary for the IBL defender, while the IBL defender can learn to counter carefully crafted optimal attack strategies. To test these observations, we conducted a third experiment, where humans played the role of defenders against both strategic and IBL attackers in an interactive task. The results confirm the predictions of the second simulation experiment: Cognitive attackers are more challenging for human defenders than strategic attackers. These insights contribute to informing future adversary emulation efforts and training of cyber defenders.

CCS Concepts: • **Computing methodologies** → **Modeling methodologies**; **Model verification and validation**; **Cognitive science**; • **Security and privacy** → **Human and societal aspects of security and privacy**; • **Human-centered computing** → **Empirical studies in interaction design;**

Additional Key Words and Phrases: Adversary emulation, cognitive modeling, cyber security game

This research was sponsored by the Army Research Office and accomplished under Australia-USA MURI Grant Number W911NF-20-S-000 and by the Army Research Laboratory under Cooperative Agreement Number W911NF-13-2-0045 (ARL Cyber Security CRA).

Authors' Contact Information: Yinuo Du, Software and Social Systems Department, Carnegie Mellon University, Pittsburgh, Pennsylvania, United States; e-mail: yinuod@andrew.cmu.edu; Baptiste Prebot, Department of Social and Decision Sciences, Carnegie Mellon University, Pittsburgh, Pennsylvania, United States; e-mail: baptiste.prebot@ensc.fr; Tyler Malloy, Department of Social and Decision Sciences, Carnegie Mellon University, Pittsburgh, Pennsylvania, United States; e-mail: tylermal@andrew.cmu.edu; Cleotilde Gonzalez, Department of Social and Decision Sciences, Carnegie Mellon University, Pittsburgh, Pennsylvania, United States; e-mail: coty@cmu.edu.



This work is licensed under a Creative Commons Attribution 4.0 International License.

© 2025 Copyright held by the owner/author(s).
ACM 2469-7818/2025/03-ART7
https://doi.org/10.1145/3712672

ACM Reference Format:
Yinuo Du, Baptiste Prebot, Tyler Malloy, and Cleotilde Gonzalez. 2025. A Cyber-War Between Bots: Cognitive Attackers are More Challenging for Defenders than Strategic Attackers. *ACM Trans. Soc. Comput.* 8, 3-4, Article 7 (March 2025), 22 pages. https://doi.org/10.1145/3712672

1 Introduction

Cyber systems have gradually populated all the personal and collective layers of society. From banks to hospitals, from electric grids to industrial facilities, the interconnectivity of systems has created new opportunities for criminals. Cyber security is a domain of great complexity, defined by uncertainty, lack of visibility, extreme speeds, and partial information. In this adversarial context, defenders and attackers confront each other using digital weapons that are beyond the limits of human capabilities for perception and assessment. Defenders need extensive experience to effectively defend against dynamic and distributed attacks.

Cyber wargaming and adversary emulation (i.e., Red teams) are common practices in organizations to train defenders (i.e., Blue teams) and to develop appropriate defense algorithms [10, 15]. However, the design of emulated adversaries can be expensive and time-consuming, especially for scaled networks with a large attack surface and rich defense arsenals. Autonomous agents have been developed to mitigate this problem [5, 47, 51]. For example, Kotenko [30] modeled a DDoS attack, and Razak et al. [43] simulated network intrusions. However, these simulations do not specifically portray the attacker. The early models contained static patterns prescribed for the attacker agents to follow [22]. These models eventually gave way to graph-based [29] and state-based [1] attack simulation methods, which provide a useful characterization of the attacker's profile, such as goals, starting points, and available time. This group of simulation methods models and stores generic attack patterns with preconditions and postconditions in a knowledge base. Additional attack pattern attributes include the cost of attempts, execution time, base success probability, and maximum attempts.

Despite technical fidelity, most automated adversary simulation methods ignored the social context and lacked a dynamic behavior component [28]. Human attackers have varying levels of risk tolerance, which might affect their choice of target and attack methods [55]. Human attackers can also learn from their experiences [31], dynamically adapt to defenses they encounter, and modify their strategies accordingly, making them more dangerous over time as they become more adept at evading detection and exploiting vulnerabilities. Thus, to improve the training of defenders, the emulated adversaries need to exhibit behavior similar to that of the human attackers and have the capability to learn and adapt to the defender's actions.

Cognitive architectures and theories of human decision-making have made significant progress in emulating human-like behavior in dynamic environments. Unlike typical computational algorithms that aim to make optimal decisions, cognitive architectures adhere to human constraints such as forgetting, limited attention, and bounded rationality [16]. Cognitive models based on **instance-based learning theory (IBLT)** [20] have been implemented in the context of cybersecurity to model human cognitive processes. Dutt et al. [13] proposed an IBL model that represents the awareness of cyber situation of a human analyst and is capable of making concrete predictions about the recognition and comprehension processes of a security expert in a cyber attack. A more recent model from Du et al. [12] uses a cyber security scenario in which the IBL defender learns to defeat the most aggressive, optimal, but deterministic attack strategy. Cognitive models have also been used as embedded computational agents to simulate human interactions with software and networks [18, 35, 52]. Previous work has focused independently on

understanding defense behaviors and developing cognitive models of blue agents [12, 13] or the attack preferences and biases of the attacker [11]. However, past work on cognitive modeling in cyber security systems has rarely considered the real-time social interactions of attackers and defenders together.

The attacker and the defender can influence each other in cyber adversarial scenarios [53]. Such dynamics between attackers and defenders can make defenders more vulnerable to adversarial actions compared to even random attackers [36]. For example, in a simple and abstract game, humans were found to handle random attacks more effectively than adaptive attacks [36]. This suggests that commonly used random attack security algorithms may be less effective than human-inspired adaptive attack strategies in training human defenders. The characteristics of human attackers have been studied in a phishing experiment. For example Rajivan and Gonzalez [42] found that individual creativity is a predictor of an adversary's ability to evade detection. Cognitive biases and emotions are also believed to affect attacker behavior and decision-making [14, 26].

Given the current evidence on cyber wargaming and adversary emulation, we hypothesize that cognitive agents that are capable of emulating *human* adversaries will be more challenging for cyber defenders than deterministic attacker strategies. If this hypothesis is correct, then training cyber defenders against cognitive adversaries will result in better-prepared defenders than the current procedure for training cyber defenders against strategic attackers. This article aims to test this hypothesis.

The contributions of this article are as follows: First, we propose a cognitive model of a red agent that uses the theoretical principles of IBLT (that is, IBL_{Red}) [20]. In a simulation experiment, we compare the performance of the IBL_{Red} agent with that of a deterministic, highly accurate, and targeted attack strategy ($Beeline_{Red}$) and with a strategy that explores the network without prior knowledge about the location of targets ($Meander_{Red}$). In Experiment 1, we demonstrate that IBL_{Red} is capable of learning and achieving performance similar to the optimal agent $Beeline_{Red}$. In Experiment 2, we tested the three red agents (IBL_{Red},$Beeline_{Red}$, and $Meander_{Red}$) against a cognitive defender IBL_{Blue} to demonstrate that IBL_{Red} is the most challenging attacker for IBL_{Blue}. In Experiment 3, we validate the simulation findings of Experiment 2 in an experiment in which human participants play the role of defender. The results of the experiment confirm that cognitive adversaries are more challenging to human defenders than strategic adversaries.

2 Instance-based Learning Theory

IBLT is a cognitive theory of decision-making. It is based on the idea that decisions are made by recognizing similar past experiences, integrating them into generating the expected utility of decision alternatives, and selecting the alternative with the maximum expected utility [20]. The development of cognitive models for cyber defense is based on a large body of work on applying cognitive science to cyber security (e.g., Reference [17]).

Although both the process and the mechanisms of IBLT have been published, we repeat the mathematical formulations of the theory here for completeness. The central element of IBLT is the "instance." It represents a unit of memory resulting from evaluating potential choice alternatives. Each decision is stored in an instance, structured with three elements that are built over time: A situation state s that is composed of a set of characteristics f; a decision or action a taken corresponding to an alternative in state s; and an expected utility or experienced result x of the action taken in a state. Concretely, for an IBL agent, an option $k = (s, a)$ is defined by action a in state s. At time t, assume that n_{kt} different instances (k_i, x_{ik_it}) for $i = 1, \ldots, n_{kt}$, associated with k. Each instance i in memory has an *activation* value, which represents the ease of retrieving this information from memory [4]. Here, we consider a simplified version of the activation equation,

which only captures recency, frequency, and noise in memory:

$$\Lambda_{ik_it} = \ln\left(\sum_{t' \in T_{ik_it}} (t - t')^{-d}\right) + \sigma \ln \frac{1-\xi_{ik_it}}{\xi_{ik_it}}, \tag{1}$$

where d and σ are the decay and noise parameters, respectively, and $T_{ik_it} \subset \{0, \ldots, t-1\}$ is the set of previous timestamps in which instance i was observed. The rightmost term represents noise to capture individual variation in activation, and ξ_{ik_it} is a random number drawn from a uniform distribution $U(0, 1)$ at each step and for each instance and option.

Activation of an instance i is used to determine the probability of retrieving an instance from memory. The probability of an instance where a soft-max function defines i:

$$P_{ik_it} = \frac{e^{\Lambda_{ik_it}/\tau}}{\sum_{j=1}^{n_{kt}} e^{\Lambda_{jk_jt}/\tau}}, \tag{2}$$

where τ is the Boltzmann constant (i.e., the "temperature") in the Boltzmann distribution. For simplicity, τ is often defined as a function of the same σ used in the activation equation $\tau = \sigma\sqrt{2}$.

The expected utility of option k is calculated based on *Blending* as specified in discrete choice tasks [19]:

$$V_{kt} = \sum_{i=1}^{n_{kt}} P_{ik_it} x_{ik_it}. \tag{3}$$

The choice rule is to select the option corresponding to the maximum blended value. When the agent receives delayed results, the agent updates the expected utilities using a credit assignment mechanism [38].

3 Cyber Security Scenario

Testing attacker and defender agents requires a simulation or training platform that encapsulates cyber elements in an integrated environment. On such a platform, defense agents can confront attack agents in cyber scenarios and network simulations. Here, we use the interactive defense game,[1] based on CybORG AI gym [7, 8, 50] with adversarial cyber-operation scenarios to allow users to train agents in a simple but realistic environment. We adopt the CAGE cyber defense scenario [49] to perform experimental simulations using IBL agents as cyber defenders and cyber attackers. This framework was also presented in References [12, 41], and in the following, we outline its main structural elements and the particularities of the cyber defense scenario.

The attacker (hereafter the Red agent) interacts with the environment through high-level actions that aim to progress and impact the network; the defender (hereafter the Blue agent) aims to stop the progression of the attacker and remove it from the network. Each combat between an attacker and a defender is an episode, and the duration of episode was set to 25 steps to ensure that the Blue agent could fully observe the attack strategies.

Figure 1 illustrates the topology of the network chosen for this scenario. The network is divided into three subnets: Subnet 1 consists of user hosts that are not critical, Subnet 2 consists of enterprise servers designed to support the user activities on Subnet 1, and Subnet 3 contains the critical operational server and operational hosts. Attackers usually start with user subnets and establish their entry point through social engineering or spear phishing. In this scenario, the entry point is host *User0*.

Figure 2 summarizes the phases of a targeted attack led by the Red agent (red arrows) and countermeasures for the Blue agent to stop it (blue arrows). The Red agent starts by searching

[1] http://janus.hss.cmu.edu:8084/

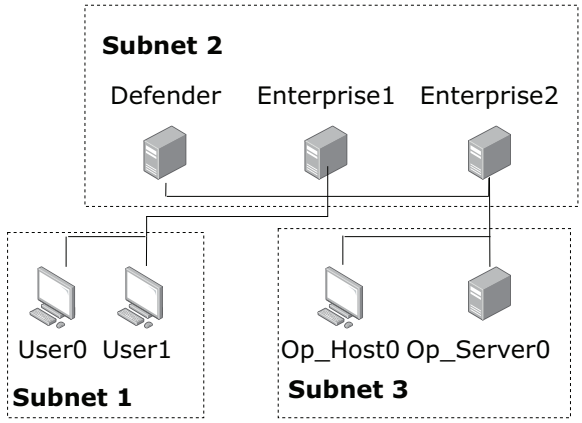


Fig. 1. Adaptation of the cage challenge network.

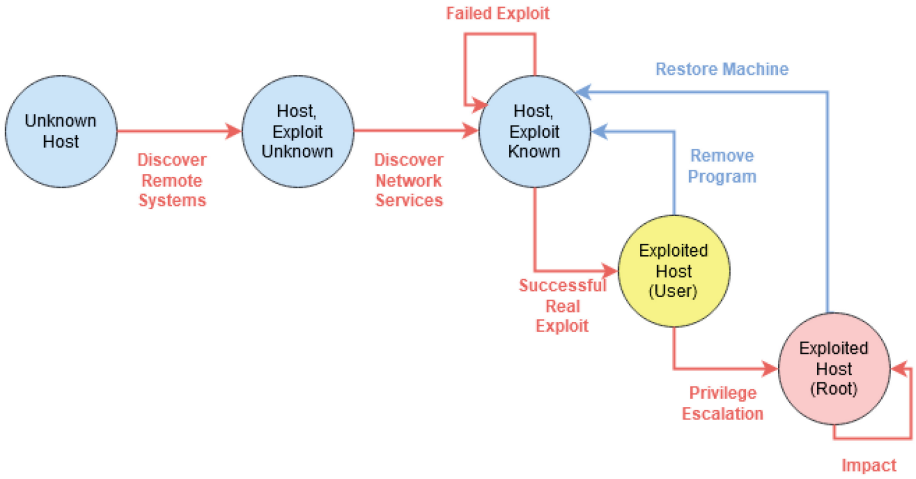


Fig. 2. Effect of actions on the host state (diagram from Reference [49]).

for hosts on the network with *DiscoverRemoteSystems*. To identify vulnerabilities on a target host, the next step is to *DiscoverNetworkServices*. A successful *ExploitRemoteService* on target can obtain *User* level access for the Red agent, which can be escalated to a more privileged *Root* level by *PrivilegeEscalate*. The Blue agent can *Remove* its adversary at the *User* level and use *Restore* if the Red agent has escalated. It can also *Analyse* the activities for additional information or passively *Monitor* the network.

3.1 Red Agents

We used three types of red agents: Two strategic attackers: (1) a highly efficient deterministic agent, $Beeline_{Red}$, and (2) a stochastic agent, $Meander_{Red}$; and (3) a dynamic cognitive agent, IBL_{Red}.

Advanced attackers well-funded by organizations or governments are well planned and highly organized to increase the probability of the success of the attack [3]. This type of attackers extensively research their target, collecting the necessary information and intelligence on the assets of the organizations. Before launching attacks, the attackers can acquire details of the

network layout, such as the types of switches, routers, anti-virus tools, firewalls, Web servers used, and ports open. The attackers can then build attack plans using well-known vulnerability databases such as the **Common Vulnerabilities and Exposures List (CVE)** and the NIST **National Vulnerability Database (NVD)** [34]. These plans allow attackers not only to establish a foothold, but also to penetrate deeper into the target's network. $Beeline_{Red}$ assumes that the attacker has prior knowledge of the network topology and moves directly to the operational server following the red path ($User0 \rightarrow User1 \rightarrow Enterprise1 \rightarrow Enterprise2 \rightarrow Op_Server0$) (see Figure 1) in a predictive and deterministic way.

Novice attackers (also known as "Script Kiddies") who rely on pre-made exploit programs and files ("scripts") are usually not dedicated enough to their hacking [21]. Instead of making careful plans and collecting the necessary tools beforehand, they tend to scan random IP blocks on the Internet for weaknesses and exploit them as they are found. With the increasing amount of tools and scripts available on the Internet for free, novice attackers can also do a lot of damage to well-protected systems [9]. $Meander_{Red}$ assumes no prior knowledge about the network structure and behaves in a stochastic manner by choosing a random target to move forward.

In contrast, the cognitive agent, IBL_{Red}, is a novel contribution to this research, and it is a dynamic agent that learns from experience, as described in the IBLT section above. IBL_{Red} intends to represent cognitive memory-based decisions that can adapt their actions dynamically according to the conditions of the environment and the actions of the blue agent. The instances represent each decision made and are structured with the following three elements:

State, s_a: The state of the instances of the IBL_{Red} agent is composed of features, f, constructed using the concept of Attack Models and Attack Graphs introduced by Sheyner et al. [48] to model the security vulnerabilities of a network and their exploitation from the perspective of an attacker. Specifically, contextual characteristics include the success status of the previous action of the IBL_{Red} agent and the resources it occupied. A slot is dedicated to each type of resource in various states, as shown in Figure 2.

Specifically, a subnet can be newly *Detected* or already *Scanned*, while hosts are classified as *Detected*, *Scanned*, *Exploited (User)*, *Exploited (Root)*, *Impacted*.

The starting status denotes when the IBL_{Red} agent has just successfully established its foothold on the network on *User0*. At that point, only the *User* subnet is detected in addition to its entry point *User0*, while the rest of the slots are empty. The most successful final state for the IBL_{Red} agent is where all hosts and servers are exploited at the *Root* level and when critical *Op_Server0* is impacted.

Action Space, a_a: The action space for the IBL_{Red} agent is dynamically constructed at each step based on the status of each host on the network. Each action consists of a target host and an applicable command. As shown in Figure 2, IBL_{Red} can choose to collect more information about hosts in the network or advance the attack status of known hosts.

Utility, z_a: A reward is calculated at each step, based on the attack status, as shown in Table 1. Higher rewards are assigned when the IBL_{Red} agent is able to access more significant systems. Only root access to the systems and successful impact on the operational server are rewarded. The agent IBL_{Red} receives a reward of 0 for any other action.

3.1.1 Metrics for the Red Agent. The performance of the Red agent was evaluated for each episode using the following metrics: **(1) Reward:** the cumulative rewards received during the execution of the scenario; **(2) Impact duration:** the average number of steps per episode that the Red agent successfully impacts the operational server; **(3) Progress:** the average number of steps

Table 1. Utility in the IBL Model: Events and Actions Costs

Event or Action	Reward
Administrator access on a Host	0.1
Administrator access on a Server	1
Successfully *Impact Op_Server0*	10

per episode that the Red agent took to penetrate *Enterprise subnet* and *Operational subnet*; and **(4) Action frequency:** the average proportion of command usage at each step in an episode.

3.2 Blue Agents

We used two types of simulated blue agents: (1) a deterministic agent $Sleepy_{Blue}$ and (2) a dynamic cognitive agent IBL_{Blue}. $Sleepy_{Blue}$ does not attempt to stop the Red agent strategically and only takes the *Monitor* action to passively observe the state of the network. This agent helps simulate the situation where the defender fails to detect the existence of the stealthy attacker and thus does not employ defense measures. IBL_{Blue} is a dynamic agent proposed and tested in Du et al. [12]. It is also based on IBLT and has been demonstrated to resemble human-like defense decisions in empirical studies [40]. The instances represent each decision made and are structured with the following three elements:

State, s_d: IBL_{Blue} instance states are constructed to resemble the information that would be presented to a human defender in the scenario. Specifically, there are two slots for each host or server, representing the observed activity and the known compromised status of that host at a certain step in an episode. The order of *(Activity, Compromised Status)* pairs for each host is fixed to encode the identity of each host, i.e., the *host name*, *IP address*, and *Subnet*. The *step index* slot is included to resemble the step counter within each episode. The IBL agent has to choose the host to protect and the tool with which to protect it. Each action consists of a host and a command in the format of *cmd host*.

Action Space, a_d: At each step, based on the observed state of the network and the consequences of the attacker's previous actions, IBL_{Blue} selects a host or server to act on and one of four possible actions: *Analyze* is used to collect information on the level of compromise of the selected host; *Remove* is used to remove a suspected malicious agent from the host or server; if the malicious agent cannot be removed, then the blue agent can *Restore* a host or server to a previous stable state; and *Monitor* to just continue observing the system, which has essentially no effect on the state of the network.

Utility, z_d: The utility of the blue agent is the negative of the utility of the red agent, as the game is zero-sum.

3.2.1 Metrics for the Blue Agent. The performance of the Blue agent was also evaluated in terms of: **(1) Action frequency:** proportions of command usages at each step; and **(2) Number of options:** the average number of defense choices available to the blue agent. Each option is a tuple that contains the command and the target host.

4 Simulation Experiment 1: Red Agents against a Deterministic Defender, $Sleepy_{Blue}$

How well do the two strategic attackers ($Beeline_{Red}$, $Meander_{Red}$) compared to the cognitive red agent (IBL_{Red}) perform against a passive defender ($Sleepy_{Blue}$)? The answer to this question provides a baseline for comparing the capabilities of the red agents.

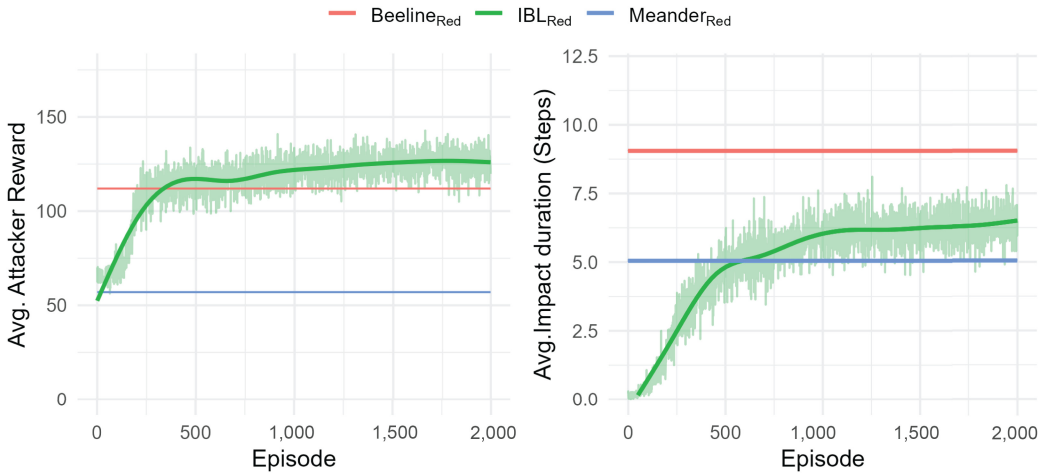


Fig. 3. Red agents performance when confronted by $Sleepy_{Blue}$: average reward per episode (left) and average impact duration (right).

Given that $Beeline_{Red}$ is the best-known strategy in our scenario, we expect that this agent would perform better compared to the other two attackers ($Meander_{Red}$ and IBL_{Red}). The $Beeline_{Red}$ agent will consistently receive the highest reward and maximum impact on the operational server, since $Beeline_{Red}$ represents a highly effective but deterministic attacker. We also expect that the $Meander_{Red}$ agent will consistently perform poorly against $Sleepy_{Blue}$, since this red agent behaves stochastically, without strategic knowledge of the path to move forward against the defender. In contrast, we expect that our newly proposed IBL_{Red} agent will initially perform poorly and similarly to $Meander_{Red}$ when confronted with the $Sleepy_{Blue}$ defender, but the IBL_{Red} agent would **learn** to take advantage of the ineffective $Sleepy_{Blue}$ defender with practice and achieve a level of performance comparable to the optimal and strategic $Beeline_{Red}$ agent.

4.1 Methods

For each type of red agent (i.e., $Beeline_{Red}$, $Meander_{Red}$, and IBL_{Red}), we performed 40 runs, each with 2,000 episodes. The IBL_{Red} agents were run with default decay $d = 0.5$ and noise $\sigma = 0.25$ parameters. This means that the results presented here are all *a priori* predictions of the IBL_{Red} agent against the $Sleepy_{Blue}$ defender.

4.2 Results

4.2.1 Dynamic Cognitive Agents Learn Effective Attack Strategies against Passive Defenders. The left panel of Figure 3 shows the reward obtained by the Red agents against $Sleepy_{Blue}$ defender. We observe that the reward of the $Beeline_{Red}$ agents was consistently larger than the reward of the $Meander_{Red}$ agents. Also, the reward of the IBL_{Red} agents was initially lower than the reward of the $Beeline_{Red}$ agents; but the IBL_{Red} agents learned over the course of the episodes, approaching the rewards of the $Beeline_{Red}$ agents after 2,000 episodes.

These observations are tested with one-way analysis of variance between subjects using *type of attack* as the main factor and *reward of attacker* as the dependent variable, aggregating for the first 500 and the last 500 episodes. As expected, the IBL_{Red} agents ($M = 59.09, SD = 43.78$) performed significantly worse than the $Beeline_{Red}$ agents ($M = 112.8, SD = 0$) against $Sleepy_{Blue}$ in the first 500 episodes [$F(1, 39998) = 30105, p < .001, \eta^2 = 0.43$]. In contrast, the IBL_{Red} agents ($M = 104.64$,

$SD = 38.68$) were able to achieve an average performance comparable to that of the $Beeline_{Red}$ agents ($M = 112.8$, $SD = 0$) in the last 500 episodes. At the end of the 2,000th episode, 55% of the agents IBL_{Red} received a higher reward than $Beeline_{Red}$, which requires them to quickly penetrate the network to impact *Op_Server0*, and at the same time fully exploit the remaining valuable systems when the opportunity arises. Most importantly, the agent IBL_{Red} learned such a complex and efficient strategy purely from experience according to IBLT [20] and without any explicit encoding of any strategy.

4.2.2 Impact Duration. The main goal of the attacker is to maintain constant *Impact* over *Op_Server0*. The right panel of Figure 3 shows the number of successive impacts performed by the red agent on the *Op_Server* after the first impact is achieved. As observed, the $Beeline_{Red}$ agents consistently maintain a longer impact duration than the $Meander_{Red}$ agents. The IBL_{Red} agents start poorly compared to the $Beeline_{Red}$ and $Meander_{Red}$ agents, but the IBL_{Red} agents are able to learn quickly and achieve a large number of impacts over the *Op_Server0* with more task practice.

To test these observations, we performed a one-way ANOVA between subjects using *attacker type* as the main factor and *impact duration* as the dependent variable, aggregating for the first 500 and the last 500 episodes. As expected, the IBL_{Red} agents are capable of achieving a duration of impact similar to that of the network ($M = 6.4$, $SD = 1$) as the $Beeline_{Red}$ agent ($M = 9.0$, $SD = 3.37$) when faced with $Sleepy_{Blue}$ in the last 500 episodes [$F(1, 39998) = 901.4, p < .001, \eta^2 = 0.31$].

5 Simulation Experiment 2: Red Agents against a Cognitive Defender, IBL_{Blue}

After performing a baseline analysis of our three red agents against a passive defender, we compared the three red agents against an adaptive cognitive defender, the IBL_{Blue} agent. In Experiment 2, the strategies of the three agents, $Beeline_{Red}$, $Meander_{Red}$, and $IBL_{Red}^{Trained}$, were kept the same as in Experiment 1. This means that the agent $IBL_{Red}^{Trained}$ was trained against the agent $Sleepy_{Blue}$ for 2,000 episodes and tested against the cognitive adaptive agent IBL_{Blue}.

Since the IBL_{Blue} agent can adapt to the strategies of the attackers, we expect that it will be able to learn effective defense strategies against the two static attackers: the $Beeline_{Red}$ and $Meander_{Red}$ agents. However, predictions of the performance of the IBL_{Blue} agent in defending against $IBL_{Red}^{Trained}$ are less clear.

We expect that $Beeline_{Red}$ will initially achieve a higher reward and a longer impact duration than the agents $IBL_{Red}^{Trained}$ and $Meander_{Red}$. However, we expect that the determinism and static nature of $Beeline_{Red}$ will be exploited by the learning agent IBL_{Blue}, resulting in a worse attack performance of $Beeline_{Red}$ than $IBL_{Red}^{Trained}$ with extended practice. We also expect that $Meander_{Red}$ will start with a lower reward and a shorter impact duration than $IBL_{Red}^{Trained}$. Furthermore, although $Meander_{Red}$ is stochastic, this attacker is not adaptive and is not a learning agent. For these reasons, we expect that, ultimately, $Meander_{Red}$ will be exploited and effectively stopped by IBL_{Blue}.

5.1 Methods

Similarly as in Experiment 1, we run 40 IBL_{Blue} runs for each type of red agent (that is, $Beeline_{Red}$, $Meander_{Red}$, and IBL_{Red}). The IBL agents (red and blue) were configured with default decay $d = 0.5$ and noise $\sigma = 0.25$ parameters.

5.2 Results: Attacker Behavior

5.2.1 Dynamic Cognitive Agents Learn Effective Defense Strategies against Static Attackers, but Not Dynamic Ones.

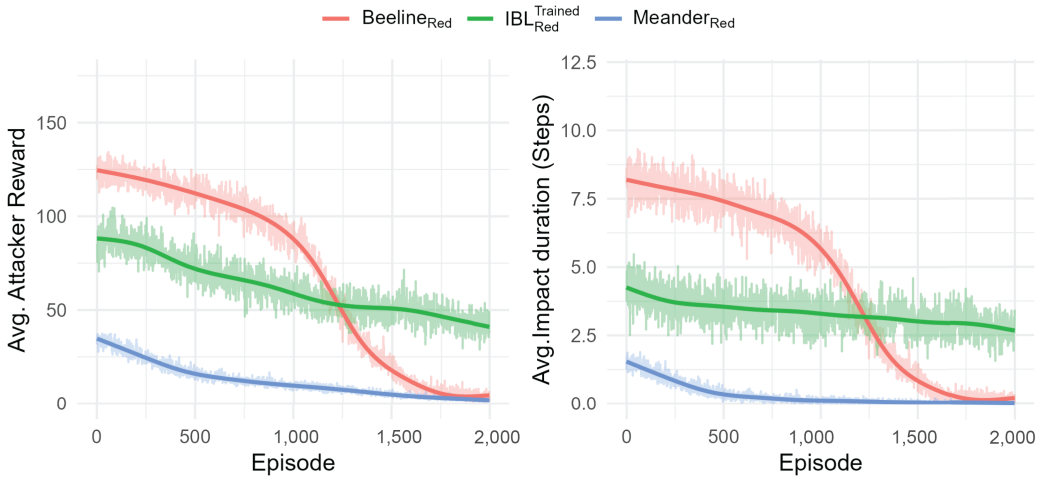


Fig. 4. Red agents performance when confronted by a cognitive defender, IBL_{Blue}. The average reward per episode (left) and average impact duration (right).

Attacker Reward. The left panel of Figure 4 shows the reward obtained by the Red agents against the IBL_{Blue} defender. We observe that the reward for the agents $Beeline_{Red}$ was initially higher than the reward for the agents IBL_{Red}. However, over time, as the agent IBL_{Blue} learns to defend the network more effectively, the reward of the agents $Beeline_{Red}$ drops significantly and approaches the reward of the agents $Meander_{Red}$. Meanwhile, the adaptive nature of the agent IBL_{Red} makes it difficult to learn a defense strategy within the training period investigated here.

Given that the agent IBL_{Blue} also starts naively learning from experience, the agents $Beeline_{Red}$ performed better ($M = 112.8$, $SD = 30.39$) than the agents $IBL^{Trained}_{Red}$ ($M = 80.34$, $SD = 45.02$) in the first 500 episodes [$F(1, 39998) = 20579, p < .001, \eta^2 = 0.34$]. However, $Beeline_{Red}$ agents ($M = 5.15$, $SD = 14.68$) performed significantly worse than $IBL^{Trained}_{Red}$ agents ($M = 54.60$, $SD = 34.31$) in the last 500 episodes [$F(1, 39998) = 31185, p < .001, \eta^2 = 0.44$]. The $Meander_{Red}$ agents ($M = 5.26$, $SD = 4.42$) start with worse performance than IBL_{Red} ($M = 80.34$, $SD = 45.02$) [$F(1, 39998) = 21185, p < .001, \eta^2 = 0.42$] and end with zero reward.

Impact duration. As shown in the right panel of Figure 4, the duration of the impact shows trends similar to the reward obtained by the red agents against IBL_{Blue}. $IBL^{Trained}_{Red}$ achieves shorter impact duration ($M = 4.08$, $SD = 3.77$) than $Beeline_{Red}$ ($M = 8.93$, $SD = 2.34$) in the first 500 episodes [$F(1, 39998) = 17240, p < .001, \eta^2 = 0.95$]. This relative disadvantage reversed in the last 500 episodes, where $IBL^{Trained}_{Red}$ had a longer impact duration ($M = 3.25$, $SD = 2.82$) than $Beeline_{Red}$: ($M = 0.18$, $SD = 1.11$) [$F(1, 39998) = 14699, p < .001, \eta^2 = 0.94$]. The agents $Meander_{Red}$ start with a worse performance ($M = 0.8711282$, $SD = 1.118891$) than IBL_{Red} ($M = 3.25$, $SD = 2.82$) [$F(1, 39998) = 19122, p < .001, \eta^2 = 0.32$] and are unable to cause an impact in the last 500 episodes.

Attacker Progress. To further explore the behavior of red agents, we analyzed the number of steps the attacker takes to reach a subnet (Enterprise and Operational). Taking into account the layered network structure shown in Figure 1, the progress of the Red agents can be marked by two milestones: penetration of the *Enterprise* subnet and the *Operational* subnet. The higher the number of steps required to reach a specific subnet, the more effective the defense agent is at protecting the network from attack.

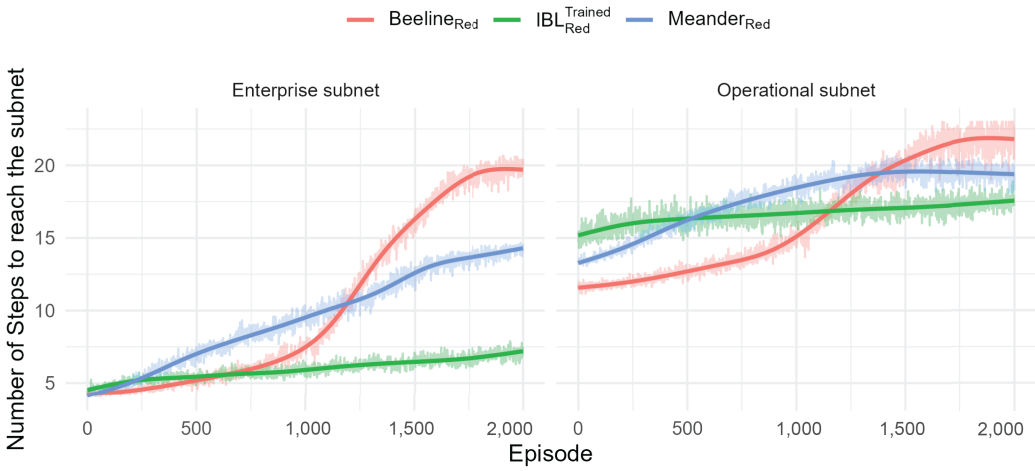


Fig. 5. Red agents progression capability when confronted by IBL_{Blue}. Number of steps taken to reach the Enterprise subnet (left) and Operational subnet (right).

We can observe the increasingly delayed and impeded forward progress of $Beeline_{Red}$ in Figure 5. In general, each red agent takes longer to access both the *Enterprise* and *Operational* subnets at the end of the training period compared to earlier in the training, demonstrating that the IBL_{Blue} agent effectively learns to defend. At the end of the training period, $Beeline_{Red}$ takes, on average, 15 more steps to enter the *Enterprise* subnet and 10 more steps to enter the *Operational* subnet compared to earlier in the training. However, $IBL^{Trained}_{Red}$ shows relatively stable rates of the number of steps taken to reach these two subnets over the course of 2,000 episodes.

In the first 500 episodes, $IBL^{Trained}_{Red}$ takes longer time to penetrate the Enterprise subnet ($M = 4.77$, $SD = 0.33$) and the Operational subnet ($M = 15.67$, $D = 0.61$) than $Beeline_{Red}$ (Enterprise ($M = 4.59$, $SD = 0.33$), Operational ($M = 12.04$, $D = 0.38$)) [Enterprise: $F(1, 39998) = 66.02, p < .001, \eta^2 = 0.06$] [Operational: $F(1, 39998) = 12489, p < .001, \eta^2 = 0..93$]. But this relative disadvantage reversed in the last 500 episodes, where the $IBL^{Trained}_{Red}$ propagated faster into the Enterprise subnet ($M = 6.12$, $SD = 0.40$) and the Operational subnet ($M = 16.90$, $D = 0.52$) than the $Beeline_{Red}$: Enterprise ($M = 19.20$, $SD = 0.99$) [$F(1, 39998) = 74265, p < .001, \eta^2 = 0.99$], Operational ($M = 21.27$, $D = 0.97$),$F(1, 39998) = 7940, p < .001, \eta^2 = 0.89$].

The faster progress speed of $IBL^{Trained}_{Red}$ emerges from its learning capability against IBL_{Blue} in addition to the stochasticity of its actions. Although $Meander_{Red}$ acts stochastically, it is significantly impeded by a lack of adaptation and takes, on average, 10 more steps to achieve each milestone than $IBL^{Trained}_{Red}$.

5.2.2 Dynamic Cognitive Agents Learn Defense Strategies That Disrupt the Behavior of Static Agents, but Not Dynamic Ones. Figure 6 compares the average distribution of the use of attack commands at each step of the first 500 episodes (left panels) versus the last 500 episodes (right panels). $IBL^{Trained}_{Red}$ presents proportions of actions similar to $Beeline_{Red}$ at the beginning, with higher *Monitor* proportion for $IBL^{Trained}_{Red}$. In the final episodes, $Beeline_{Red}$ and $Meander_{Red}$ are stuck in a loop of *ExploitRemoteService* and *PrivilegeEscalate*, while $IBL^{Trained}_{Red}$ maintained a consistent distribution. This comparison constitutes further evidence of the inefficacy of strategic attack agents. The disappearance of *Monitor*, *Analyse*, and *Impact* actions can help explain the reason for the rapid drop in reward within the episodes.

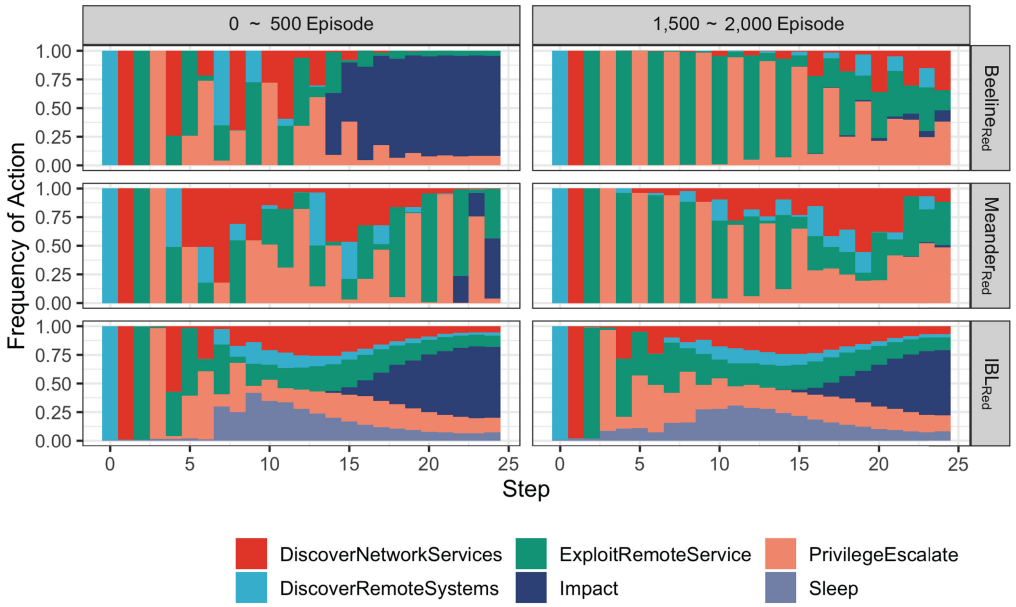


Fig. 6. Red agents evolution of action frequency in the first 500 episodes (left) and the last 500 episodes (right).

5.3 Results: Defender Agent Behavior IBL_{Blue}

Since the adversarial scenario is a zero sum game (i.e., the loss of the defender corresponds to the reward of the attacker), the performance results for IBL_{Blue} can be derived from the performance results of the red agents presented above. In particular, IBL_{Blue} performed better against $Meander_{Red}$ and $Beeline_{Red}$ than against $IBL_{Red}^{Trained}$. It was able to learn to defend effectively against $Meander_{Red}$ and $Beeline_{Red}$, eventually achieving near-zero losses in the last 500 episodes. However, IBL_{Blue} was only able to reduce the loss of attacks $IBL_{Red}^{Trained}$ by half.

To provide a deeper understanding of these results, we focus this section on the exploration of the defender's behavior against the three attackers.

5.3.1 Dynamic Cognitive Agents Adapt Defense Actions Taken against Static Agents, but Not Dynamic Ones. As presented in Figure 7, the dynamics of the use of defensive commands by the agent IBL_{Blue} shows a difference when confronting the agents $Beeline_{Red}$ and $Meander_{Red}$ in contrast to the agent $IBL_{Red}^{Trained}$. IBL_{Blue} agents faced with a strategic attacker are able to minimize the proportion of costly *Restore* action and stop the attacker with *Remove* in an earlier state of the cyber attack chain.

When the IBL_{Blue} against are defending against an $IBL_{Red}^{Trained}$ attacker, they do not adjust the actions they take in a way that is able to respond appropriately to the attacker. This makes sense, given the constantly adapting nature of the $IBL_{Red}^{Trained}$ attacker's behavior. These results indicate that the dynamics of this environment make it difficult for an adaptive defense strategy to learn effectively against an attacker that is also adapting their strategy.

5.3.2 Dynamic Cognitive Attackers Force Defenders to Choose from More Action Options Compared to Static Attackers. Psychology and behavioral research has shown that a large number

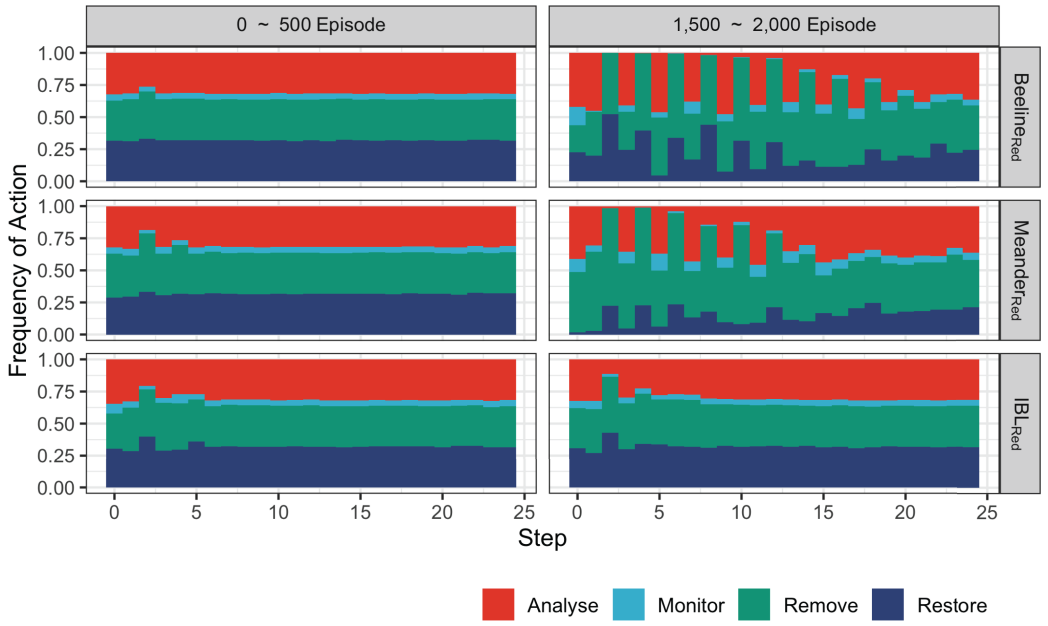


Fig. 7. IBL_{Blue} evolution of action frequency in the first 500 episodes (left) and the last 500 episodes (right).

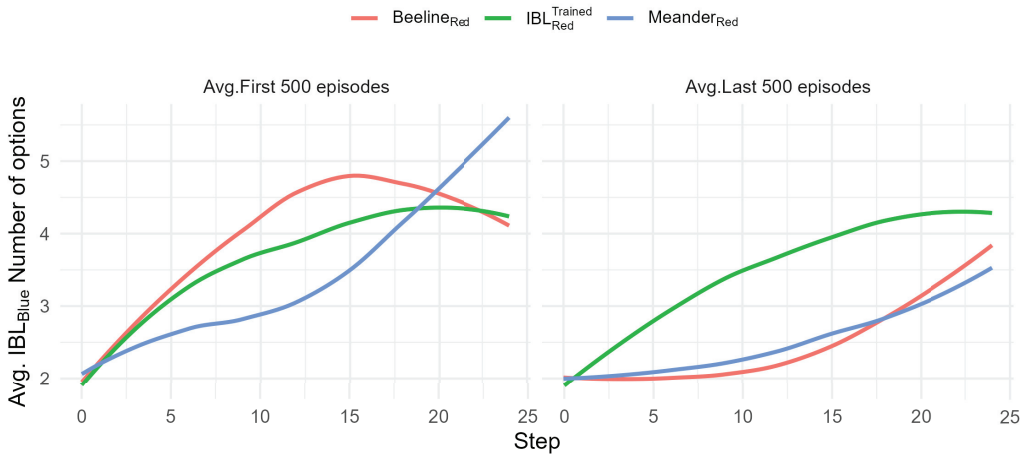


Fig. 8. Size of IBL_{Blue}'s option space in the first 500 episodes (left) and the last 500 episodes (right).

of options hinder learning speed [44, 46]. Defenders face this challenge of information overload. Figure 8 analyzes the number of options available to the IBL_{Blue} agent during the 25 steps of the episodes. As shown in the left panel, when IBL_{Blue} fought against $Beeline_{Red}$ or $Meander_{Red}$, it was able to reduce the option space in the final 500 episodes compared to the first 500 episodes. In contrast, the option space of IBL_{Blue} stays about the same size from the first 500 episodes to the last 500 episodes when facing the adaptive agent $IBL_{Red}^{Trained}$. That is, agent IBL_{Blue} was unable to simplify its option space by impeding its progress and minimizing the number of attacked hosts.

In summary, IBL_{Blue} is able to win the battle against $Beeline_{Red}$ and $Meander_{Red}$ after 2,000 episodes of repetitive interactions. It is able to block the opponent's progress, reduce the cognitive load of himself, and eliminate loss. $IBL_{Red}^{Trained}$ appears to be much more persistent, and IBL_{Blue} can only slightly alleviate its impact.

6 Experiment 3: Red Agents against Human Defenders

Although IBL models have shown successful replication of human behaviors in a variety of tasks, including cyberdefense [17], empirical verification of the findings of Experiment 2 with human defenders is needed. The first goal of this experiment is to compare the performance of human defenders faced with the three types of attackers, $Beeline_{Red}$, $Meander_{Red}$, and $IBL_{Red}^{Trained}$. The second goal of this experiment is to validate the simulation results of predicted human behavior when paired against these three attackers, by comparing human performance with simulated IBL_{Blue} defenders. The results of this experimentation will provide support for the theoretical contribution of this work in motivating the use of cognitive agents to train cyber defenders.

6.1 Experimental Design

Human participants completed the same cyber defense task and scenario as IBL_{Blue}, presented in Section 3. They performed the task using **Interactive Defense Game (IDG)** [40, 41], which provides an interactive decision game in the cyber task and environment.

6.2 Participants

Participants were recruited through Amazon Mechanical Turk to participate in a cybersecurity study. The study was advertised to last between 30 and 45 minutes. The time it took between participants was M = 47.29± 16.36 minutes. Participants received a base compensation of $4.50, and up to $5.60 in bonus payment (M = 3.50±1.39) based on their final score.[2] 186 participants (124 men, 61 women, 1 N / A) aged 21 to 65 years (M = 37.12 ± 10.15) completed the study. Seventeen of the 186 participants (9%) had more than five years of experience in the network operation and security area and at least a Master's degree in a related field. Each participant was randomly assigned to face one of the three red agents: $Beeline_{Red}$, $Meander_{Red}$, or $IBL_{Red}^{Trained}$.

6.3 Procedure

After giving their informed consent and completing a demographic questionnaire, the participants received instructions for the task followed by a short quiz to verify their basic understanding of the instructions for the task. The participants had to correctly answer all the questions before moving on to the next step of the experiment. The participants received feedback on the precision of their responses. There was no limit in the number of attempts the participants had to answer the questions correctly. However, we recorded the score of their first attempt and the number of times they tried to answer the questions. The participants then watched a video introduction to the IDG explaining the interface, the game controls, and the dynamics of an episode.

The participants then performed the task consisting of two phases: (1) a practice session and (2) a main task. The practice session consisted of two short episodes (i.e., games) of 10 steps each. The practice episodes were intended to familiarize participants with the interface and game controls. To do so, the participants successively faced Beeline and Meander; however, since these two deterministic attack strategies do not differ significantly during the first 10 steps, the participants did not have enough information to discriminate between them during the practice session.

[2] As the score used in this experiment is negative (loss), the bonus payment was calculated by using the difference to the maximum possible loss and attributing $0.005 per point: bonus=(*total loss*+1120)*0.005.

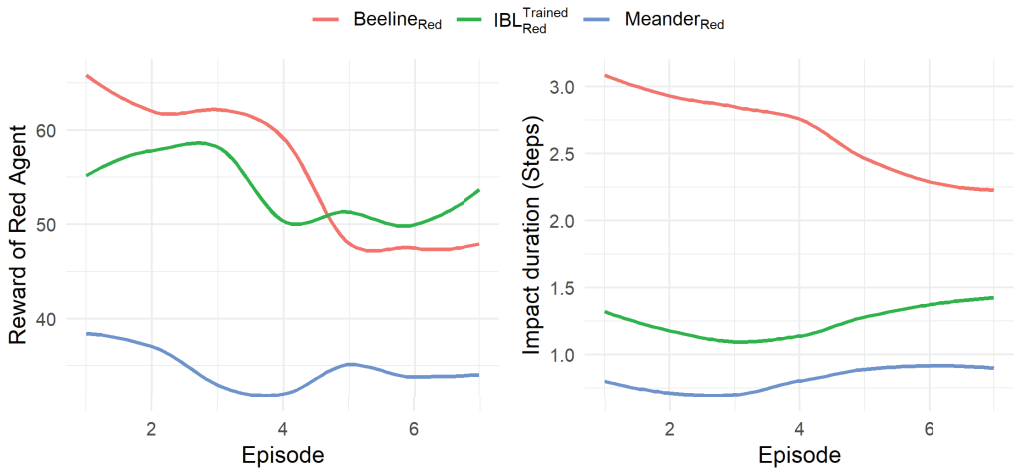


Fig. 9. Red agents performance when confronted by a human defender. The average reward per episode (left) and average impact duration (right).

Following the practice session, the participants performed the main task consisting of 7 episodes of 25 steps against the same type of adversary. No time restrictions were imposed. The initial state of the network was the same for all participants and for each of the episodes.

Subsequently, participants completed a post-experiment survey composed of two parts: (1) feedback on their performance and perceived strategy and (2) their experience in computer science and cyber defense. Finally, the participants received their final score and were dismissed. Experimental instructions, quizzes, and surveys, along with data and analysis scripts, can be accessed at https://osf.io/8vxej/?view_only=c42691c2b5bb4c31a72b1ada00e38428.

6.4 Results: Attacker Behavior against Human Defenders

6.4.1 Human Defenders Perform Similarly to Dynamic Cognitive Defenders against Both Static and Dynamic Attackers. The performance of three types of attackers ($Beeline_{Red}$, $Meander_{Red}$, $IBL_{Red}^{Trained}$) against human defenders shown in the left panel of Figure 9 is in alignment with the simulation predictions shown in Figure 4. As human participants learn from experience, the $Beeline_{Red}$ agents performed better ($M = 66.25, SD = 5.39$) than the $IBL_{Red}^{Trained}$ agents ($M = 55.51$, $SD = 4.70$) in the first episode [$F(1, 124) = 2.187, p = 0.142, \eta^2 = 0.017$].

However, $IBL_{Red}^{Trained}$ agents posed a more persistent threat to the human defender than $Beeline_{Red}$ agents. The performance of the $Beeline_{Red}$ agents deteriorates rapidly. The $Beeline_{Red}$ agents ($M = 47.33$, $SD = 6.42$) performed worse than the $IBL_{Red}^{Trained}$ agents ($M = 54.141$, $SD = 6.191$) in the last episode [$F(1, 124) = 0.577, p = 0.449, \eta^2 = 0.005$]. The $Meander_{Red}$ agents ($M = 5.26$, $SD = 4.42$) perform significantly worse [$F(1, 778) = 46.762, p < .001, \eta^2 = 0.057$] than $IBL_{Red}^{Trained}$ in 7 episodes.

Consistent with the simulation prediction shown in Figure 4, $IBL_{Red}^{Trained}$ demonstrates the persistent duration of the impact and is superior to $Meander_{Red}$ ($M = 0.817$, $SD = 1.488$). In addition, $IBL_{Red}^{Trained}$ achieves a shorter impact duration ($M = 1.29$, $SD = 0.18$) than $Beeline_{Red}$ ($M = 3.15$, $SD = 0.39$) in the first episodes [$F(1, 872) = 53.1, p < .001, \eta^2 = 0.95$]. This gap decreased significantly at the end of the seventh episode [$Beeline_{Red}$: ($M = 2.23$, $SD = 0.41$), $IBL_{Red}^{Trained}$: ($M = 1.44$, $SD = 0.35$)].

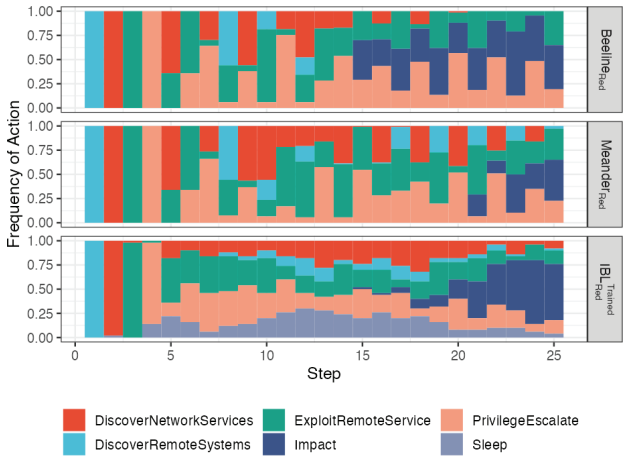


Fig. 10. Distribution of attack commands against human defenders.

These results show that the ordering of the highest and lowest attacker rewards and impact duration on early trials match between human and cognitive defenders. Additionally, the trends in adjusting the performance of attackers throughout the learning of the defender are similar. Overall, these results indicate a similar performance against all attack strategies between human and dynamic cognitive model defenders.

6.4.2 Distribution of Attack Commands. As demonstrated in Figure 10, $IBL_{Red}^{Trained}$ used the *Impact* command the most often, while $Beeline_{Red}$ and $Meander_{Red}$ are unable to exert a consistent impact on the operational server and resort to *ExploitRemoteService* and *PrivilegeEscalate*.

6.4.3 Human Participants and Cognitive Defenders Behave Similarly against All Attacker Strategies. Consistent with the predictions of the simulated human performance, there was a significant difference in the mean reward by episode based on the type of Red agent (F(2,)=28.56, p<1e-10, η^2=0.5).

6.4.4 Action Frequency. Similar to IBL_{Blue}, the use of defensive commands by the human defender shows a difference when confronting the agent $Beeline_{Red}$ and $Meander_{Red}$ in contrast to the agent $IBL_{Red}^{Trained}$. Human participants are more passive and take more *Analyse, Monitor* actions than *Remove, Restore* actions. However, as shown in Figure 11, human participants have a consistent preference for the choice of action throughout the course of 7 episodes.

6.4.5 Size of Option Space. Figure 12 presents the number of options available to the human during the 25 steps of the episodes. Similarly to IBL_{Blue}, human participants can alleviate their cognitive load by narrowing down the option space when paired with $Beeline_{Red}$. The size of the option space remains approximately the same in the two stochastic conditions, that is, $Meander_{Red}$ and $IBL_{Red}^{Trained}$.

6.4.6 Dynamic Cognitive Attackers Perform Best against the Most Efficient Human Defenders. To further investigate the human defenders against the various type of attacker models, we split the human defenders according to the distribution of attacker reward for each type of red agent. Efficient Defenders were those that resulted in attacker reward lower than the mean attack reward, and Inefficient Defenders were those that were equal or above the mean attack reward.

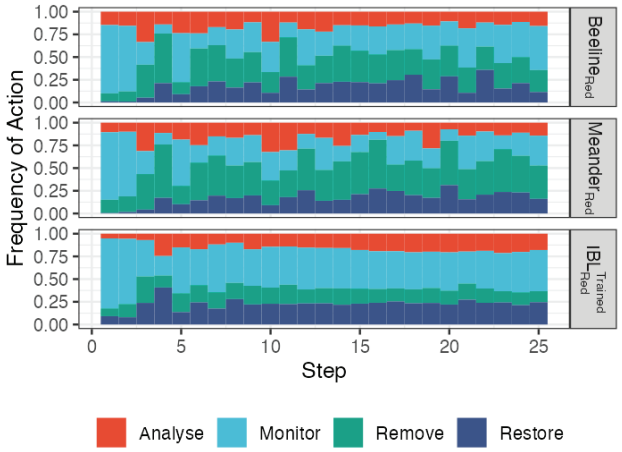


Fig. 11. Average action frequency of human defender.

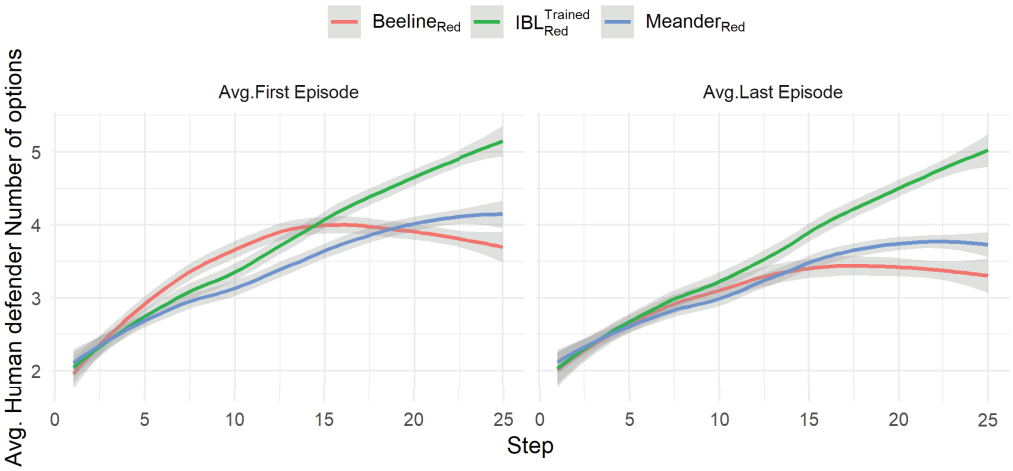


Fig. 12. Average size of the human defender's option space in the first (left) and the last episode (right).

Comparing performance in this way allowed us to determine if there was a clear difference in how red agents performed versus human participants who were better or worse able to learn attacker strategies. Furthermore, to compare performance once human defenders had enough experience to learn the attacker strategy adequately, we limited the statistical analysis to later (>4) episodes of the trial.

Figure 13 shows that $IBL_{Red}^{Trained}$ had a significantly higher reward against efficient Defenders in later trials (mean: 33.19 ± 33.31 (SD); Tukey's HSD p=0.040). Meanwhile, $Beeline_{Red}$ had a higher reward against Inefficient Defenders in later trials (mean: 92.18 ± 53.36 (SD); Tukey's HSD p=0.041). These results demonstrate that the $IBL_{Red}^{Trained}$ strategy remained a challenge for efficient Defenders throughout the experiment. Furthermore, the difficult but deterministic nature of the $Beeline_{Red}$ strategy was more difficult for inefficient Defenders.

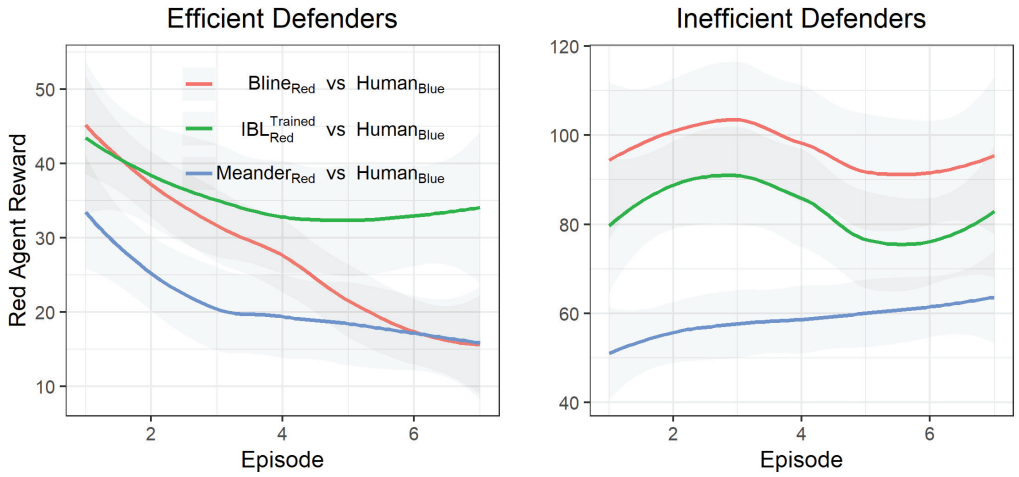


Fig. 13. Average red agent reward by episode, split between efficient human defenders and inefficient human defenders.

An explanation for the poorer performance of efficient Defenders against $IBL_{Red}^{Trained}$ and inefficient Defenders against $Beeline_{Red}$ is the defense style used by those groups. The four actions taken in the task can be described as passive (monitor and analyze) or active (remove and restore) [40]. Since $Beeline_{Red}$ quickly attacks the operational server, it could be that Inefficient Defenders perform worse against $Beeline_{Red}$. Similarly, because $IBL_{Red}^{Trained}$ is both stochastic and adaptive to defender behavior, it could be more difficult for efficient Defenders.

To test this explanation, we compared the proportion of active-type actions performed by efficient Defenders paired with $IBL_{Red}^{Trained}$, $Beeline_{Red}$, and $Meander_{Red}$. This comparison demonstrated a higher rate of active actions in efficient Defenders paired with $IBL_{Red}^{Trained}$ than both $Beeline_{Red}$ (p=0.007) and $Meander_{Red}$ (p<0.001), but no significant difference between $Beeline_{Red}$ and $Meander_{Red}$ (p=0.457). This shows that achieving proficient performance against $IBL_{Red}^{Trained}$ required more active strategic actions.

7 Discussion

Adversary emulation strategies can be used to train cyber defenders, develop intelligent systems for cyber defense, and test cyber defense capabilities. However, the process of developing effective adversary emulations can be expensive, and their evaluation is often subjective [45, 54]. Existing automated attacker strategies determine how an attacker could achieve a specific goal by assembling vulnerabilities in a graph or by executing a sequence of actions (for example, reconnaissance, escalate, exfiltrate, and lateral movement) in order [23]. Most of them are optimal, but also deterministic and predictable, and therefore can be easily thrown off by human defenders [2]. The first contribution of this article is to demonstrate that it is possible to evaluate intelligent cyber defense systems using cognitive models, aimed at emulating adaptive human decision-making [18]. We show that cognitive models that emulate human adversaries can be better test cases for cyber defenders and for technological capabilities than strategic attack agents that execute a sequence of actions in a fixed order.

Second, we present a cognitive model of an attacker based on IBLT [20], IBL_{Red}. IBL_{Red} is first trained against a static and inactive defender, $Sleep_{Blue}$. The main feature of IBL_{Red} is that it can learn from interactive feedback on the task, and we show that it can reach the same level

of effectiveness as the best adversarial strategy in this scenario. This result suggests that an IBL cognitive agent can be an effective dynamic and adaptive emulator of attack behavior. Importantly, the IBL attacker can adapt and learn according to the dynamics of the cyber defense environment. In recent years, **reinforcement learning (RL)** has also made rapid progress as an approach to building adaptive agents [33]. The difference is that RL algorithms are mostly focused on optimally solving computational problems, while cognitive agents focus more on replicating the way humans actually learn [16]. With advances in tool-supported RL agents [24] and human-autonomy teaming [39], defenders are likely to face both optimal adaptive agents and a human attacker. Thus, it is important to train against both types of adaptive attackers.

A third contribution of this article is to demonstrate the performance of an IBL model of the defender IBL_{Blue} when paired with three different types of emulated attackers: the optimal attack strategy $Beeline_{Red}$ the stochastic attack strategy $Meander_{Red}$ and a cognitive attacker $IBL_{Red}^{Trained}$ is more difficult for the IBL_{Blue} defender to learn than an optimal but stable optimal attack strategy. The explanation is that using a cognitive model to emulate attackers is more effective than using deterministic strategies. Cognitive models are dynamic and adaptive to the defender's actions, while the Beeline strategy is static and consistent. The agent IBL_{Blue} was able to learn the Beeline strategy and eventually take advantage of it, while it did not effectively hinder the progress of the agent IBL_{Red}. Our analyses show that it takes significantly more steps in time for $Beeline_{Red}$ to reach the Enterprise subnet and, ultimately, more steps to reach the Operational server. The IBL_{Blue} learns over time to prevent these actions from this $Beeline_{Red}$ strategy. However, it is significantly more difficult to prevent IBL_{Red} from reaching the Enterprise and Operational servers. We further verify that there is important learning that occurs from the first to the last episodes in terms of the actions taken by the attacker. For example, the number of impact actions is significantly reduced from the first to the last 500 episodes when the IBL_{Blue} agent confronts the $Beeline_{Red}$ strategy, but the reduction in impact actions is minimal when the IBL_{Blue} agent confronts the cognitive agent IBL_{Red}. Exploring the actions taken by the agent IBL_{Blue} suggests that the agent learns to decrease the restore actions when confronted with the agent $Beeline_{Red}$, while maintaining a more consistent distribution of actions when confronted with the agent IBL_{Red}. When analyzing the options with which the agent IBL_{Blue} is confronted at each particular time, we observed an interesting effect: The IBL_{Blue} agent learned to reduce its decision option space against $Beeline_{Red}$, while the option space of the IBL_{Blue} agent against IBL_{Red} did not decrease substantially.

Finally, we corroborate the simulation predictions in an experiment involving human defenders facing IBL_{red} and $Beeline_{Red}$ and $Meander_{Red}$. Consistent with the simulation results in IBL_{Blue}, we observe that the cognitive IBL_{Red} attacker poses a greater challenge to human defenders than the deterministic $Beeline_{Red}$ and $Meander_{Red}$ strategies. In "Interactive Computer-based" cybersecurity skill training, trainees are faced with an adversary and must take a proper course of action or face severe penalties [37]. In existing game training platforms such as CyberCIEGE [25] and CyberNEXS [6], the adversary is played by experienced hackers or emulated by automated attackers. The cognitive attacker agent can serve as another type of training partner in such interactive gaming platforms and mitigate the scarcity of human experts. Combined with techniques that trace the learning progress of trainees, the cognitive attacker agent can provide a personalized training experience for future cyber professionals.

7.1 Conclusion and Limitations

In conclusion, we provide important steps towards establishing emulated adversaries that can effectively train cyber defenders and support the development of autonomous cyber defenders. We demonstrate that it is possible to use cognitive agents to produce adversaries that are adaptive to defenders' actions. These models can ultimately be more effective in learning cyber defense

strategies than static and deterministic adversaries. In future work, we will also evaluate other types of adaptive agents (e.g., reinforcement learning–based agents) to further investigate the role of attacker adaptivity in human defender learning.

However, the cognitive agent in this work does not fully reflect the decision-making of the human attacker. In addition to learning through experience, human attackers also use various forms of fast and slow reasoning, including heuristics. Such reasoning abilities can improve attackers' efficiency but also make them vulnerable to biased decision-making. Defenders must prepare against experienced attackers who can make fast decisions with heuristics. On the flip side, there is also the potential to exploit these aspects of malicious actors, induce decision-making errors, and reduce the impacts or success of an attack. In future work, we will imbue the cognitive attacker agent with heuristic reasoning abilities. Through training against more human-like attacker agents, defenders can learn to disrupt attacker cognition.

Another limitation of this work is that the participants recruited from MTurk are not necessarily experts in cyber defense. Consistent with previous studies [27, 32], we found that defenders with different expertise have a different learning behavior. In future research, we will control the skill level of defenders by recruiting expert/novices from security operation centers and hacker communities.

Finally, demonstrating the benefits of using cognitive models in real-world cybersecurity environments remains a research challenge. Extending the scenario to the size of real-world networks can exponentially expand the state space in the cognitive model, and research on partially observable states for the defender will be required to account for imperfect network monitoring infrastructures. Future work will aim to improve the game model to be more representative of real-world environments; in particular, we will address the development of a collaborative defense environment to further explore human–AI collaboration in cyber defense.

References

[1] Robert K. Abercrombie, Bob G. Schlicher, and Frederick T. Sheldon. 2014. Security analysis of selected AMI failure scenarios using agent based game theoretic simulation. In *47th Hawaii International Conference on System Sciences*. IEEE, 2015–2024.

[2] Ron Alford, Lukas Chrpa, Mauro Vallati, and Andy Applebaum. 2022. Knowledge reformulation and deception as a defense against automated cyber adversaries. In *The International FLAIRS Conference Proceedings*, Vol. 35. The Florida AI Research Society. https://journals.flvc.org/FLAIRS/article/view/130675

[3] Adel Alshamrani, Sowmya Myneni, Ankur Chowdhary, and Dijiang Huang. 2019. A survey on advanced persistent threats: Techniques, solutions, challenges, and research opportunities. *IEEE Commun. Surv. Tutor.* 21, 2 (2019), 1851–1877.

[4] John R. Anderson and Christian J. Lebiere. 1998. *The Atomic Components of Thought*. Psychology Press, New York. 504 pages. DOI: https://doi.org/10.4324/9781315805696

[5] Andy Applebaum, Doug Miller, Blake Strom, Chris Korban, and Ross Wolf. 2016. Intelligent, automated red team emulation. In *32nd Annual Conference on Computer Security Applications*. 363–373.

[6] Tolulope Awojana and Te-Shun Chou. 2019. Overview of learning cybersecurity through game based systems. In *Conference for Industry and Education Collaboration (CIEC'19)*.

[7] Callum Baillie, Maxwell Standen, Jonathon Schwartz, Michael Docking, David Bowman, and Junae Kim. 2020. Cyborg: An autonomous cyber operations research gym. *arXiv:2002.10667* (2020).

[8] Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. 2016. OpenAI Gym. *ArXiv* abs/1606.01540 (2016).

[9] Samuel Chng, Han Yu Lu, Ayush Kumar, and David Yau. 2022. Hacker types, motivations and strategies: A comprehensive framework. *Comput. Hum. Behav. Rep.* 5 (2022), 100167.

[10] Edward J. M. Colbert, Alexander Kott, and Lawrence P. Knachel. 2020. The game-theoretic model and experimental investigation of cyber wargaming. *J. Defense Model. Simul.* 17, 1 (2020), 21–38.

[11] Edward A. Cranford, Cleotilde Gonzalez, Palvi Aggarwal, Sarah Cooney, Milind Tambe, and Christian Lebiere. 2020. Toward personalized deceptive signaling for cyber defense using cognitive models. *Topics Cogn. Sci.* 12, 3 (2020), 992–1011.

[12] Yinuo Du, Baptiste Prébot, Xiaoli Xi, and Cleotilde Gonzalez. 2022. Towards autonomous cyber defense: Predictions from a cognitive model. In *Human Factors and Ergonomics Society Annual Meeting*.
[13] Varun Dutt, Young-Suk Ahn, and Cleotilde Gonzalez. 2011. Cyber situation awareness: Modeling the security analyst in a cyber-attack scenario through instance-based learning. In *Data and Applications Security and Privacy XXV*, Yingjiu Li (Ed.). Springer Berlin, 280–292.
[14] Kimberly Ferguson-Walter, Robert Gutzwiller, Dakota Scott, and Craig Johnson. 2021. Oppositional human factors in cybersecurity: A preliminary analysis of affective states. DOI: https://doi.org/10.1109/ASEW52652.2021.00040
[15] Kimberly Ferguson-Walter, Temmie Shade, Andrew Rogers, Michael Christopher Stefan Trumbo, Kevin S. Nauer, Kristin Marie Divis, Aaron Jones, Angela Combs, and Robert G. Abbott. 2018. *The Tularosa Study: An Experimental Design and Implementation to Quantify the Effectiveness of Cyber Deception*. Technical Report. Sandia National Lab, Albuquerque, NM (United States).
[16] Cleotilde Gonzalez. 2024. Building human-like artificial agents: A general cognitive algorithm for emulating human decision-making in dynamic environments. *Perspectives on Psychological Science* 19, 5 (2024), 860–873.
[17] Cleotilde Gonzalez, Palvi Aggarwal, Christian Lebiere, and Edward Cranford. 2020. Design of dynamic and personalized deception: A research framework and new insights. In *53rd Hawaii International Conference on System Sciences*. 1825–1834.
[18] Cleotilde Gonzalez, Noam Ben-Asher, Alessandro Oltramari, and Christian Lebiere. 2014. *Cognition and Technology*. Springer International Publishing, Cham, 93–117. DOI: https://doi.org/10.1007/978-3-319-11391-3_6
[19] Cleotilde Gonzalez and Varun Dutt. 2011. Instance-based learning: Integrating sampling and repeated decisions from experience. *Psychol. Rev.* 118, 4 (2011), 523.
[20] Cleotilde Gonzalez, Javier F. Lerch, and Christian Lebiere. 2003. Instance-based learning in dynamic decision making. *Cogn. Sci.* 27, 4 (2003), 591–635.
[21] Sara L. N. Hald and Jens M. Pedersen. 2012. An updated taxonomy for characterizing hackers according to their threat properties. In *14th International Conference on Advanced Communication Technology (ICACT'12)*. IEEE, 81–86.
[22] Samuel N. Hamilton and Wendy L. Hamilton. 2008. Adversary modeling and simulation in cyber warfare. In *IFIP International Information Security Conference*. Springer, 461–475.
[23] Jörg Hoffmann. 2015. Simulated penetration testing: From "Dijkstra" to "Turing test++." In *International Conference on Automated Planning and Scheduling*. 364–372.
[24] Matthias Hutsebaut-Buysse, Kevin Mets, and Steven Latré. 2022. Hierarchical reinforcement learning: A survey and open research challenges. *Mach. Learn. Knowl. Extract.* 4, 1 (2022), 172–221.
[25] Cynthia E. Irvine, Michael F. Thompson, and Ken Allen. 2005. CyberCIEGE: Gaming for information assurance. *IEEE Secur. Privac.* 3, 3 (2005), 61–64.
[26] Chelsea K. Johnson, Robert S. Gutzwiller, Joseph Gervais, and Kimberly J. Ferguson-Walter. 2021. Decision-making biases and cyber attackers. In *36th IEEE/ACM International Conference on Automated Software Engineering Workshops (ASEW'21)*. IEEE, 140–144.
[27] Keith S. Jones, Akbar Siami Namin, and Miriam E. Armstrong. 2018. The core cyber-defense knowledge, skills, and abilities that cybersecurity students should learn in school: Results from interviews with cybersecurity professionals. *ACM Trans. Comput. Educ.* 18, 3 (2018), 1–12.
[28] Hamdi Kavak, Jose J. Padilla, Daniele Vernon-Bido, Saikou Y. Diallo, Ross Gore, and Sachin Shetty. 2021. Simulation for cybersecurity: State of the art and future directions. *J. Cybersecur.* 7, 1 (2021), tyab005.
[29] Elmar Kiesling, Christine Strauss, Andreas Ekelhart, Bernhard Grill, and Christian Stummer. 2013. Simulation-based optimization of information security controls: An adversary-centric approach. In *Winter Simulations Conference (WSC'13)*. IEEE, 2054–2065.
[30] Igor Kotenko. 2007. Multi-agent modelling and simulation of cyber-attacks and cyber-defense for homeland security. In *4th IEEE Workshop on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications*. IEEE, 614–619.
[31] Pat Langley. 2022. The computational gauntlet of human-like learning. In *AAAI Conference on Artificial Intelligence*. 12268–12273.
[32] Clemens M. Lechner, Daniel Danner, Fabian Flöck, Lisa Posch, Arnim Bleier, and Markus Strohmaier. 2019. Measuring motivations of crowdworkers: The multidimensional crowdworker motivation scale. *ACM Transactions on Social Computing* 2, 2 (2019), 1–34.
[33] Yutaka Matsuo, Yann LeCun, Maneesh Sahani, Doina Precup, David Silver, Masashi Sugiyama, Eiji Uchibe, and Jun Morimoto. 2022. Deep learning, reinforcement learning, and world models. *Neural Netw.* 152 (2022), 267–275.
[34] Peter Mell, Karen Scarfone, and Sasha Romanosky. 2006. Common vulnerability scoring system. *IEEE Secur. Privac.* 4, 6 (2006), 85–89.

[35] Konstantinos Mitsopoulos, Sterling Somers, Joel Schooler, Christian Lebiere, Peter Pirolli, and Robert Thomson. 2021. Toward a psychology of deep reinforcement learning agents using a cognitive architecture. *Topics in Cognitive Science* 14, 4 (2021), 756–79.

[36] Frédéric Moisan and Cleotilde Gonzalez. 2017. Security under uncertainty: Adaptive attackers are more challenging to human defenders than random attackers. *Front. Psychol.* 8 (2017), 982.

[37] Ajay Nagarajan, Jan M. Allbeck, Arun Sood, and Terry L. Janssen. 2012. Exploring game design for cybersecurity training. In *IEEE International Conference on Cyber Technology in Automation, Control, and Intelligent Systems (CYBER'12)*. IEEE, 256–262.

[38] Thuy Ngoc Nguyen, Chase McDonald, and Cleotilde Gonzalez. 2021. *Credit Assignment: Challenges and Opportunities in Developing Human-like AI Agents*. Technical Report. Carnegie Mellon University.

[39] Thomas O'Neill, Nathan McNeese, Amy Barron, and Beau Schelble. 2022. Human–autonomy teaming: A review and analysis of the empirical literature. *Hum. Factors* 64, 5 (2022), 904–938.

[40] Baptiste Prebot, Yinuo Du, and Cleotilde Gonzalez. 2023. Learning about simulated adversaries from human defenders using interactive cyber-defense games. arXiv:2304.01142 [cs.CR]

[41] Baptiste Prébot, Yinuo Du, Xiaoli Xi, and Cleotilde Gonzalez. 2022. Cognitive models of dynamic decision in autonomous intelligent cyber defense. In *2nd International Conference on Autonomous Intelligent Cyber-defence Agents (AICA'22)*.

[42] Prashanth Rajivan and Cleotilde Gonzalez. 2018. Creative persuasion: A study on adversarial behaviors and strategies in phishing attacks. *Front. Psychol.* 9 (2018), 135.

[43] Shabana Razak, Mian Zhou, and Sheau-Dong Lang. 2002. Network intrusion simulation using OPNET. In *OPNETWORKS Conference*. Citeseer.

[44] Derek D. Reed, Brent A. Kaplan, and Adam T. Brewer. 2012. Discounting the freedom to choose: Implications for the paradox of choice. *Behav. Process.* 90, 3 (2012), 424–427.

[45] Lorenzo Russo, Francesco Binaschi, Alessio De Angelis, A. Armando, M. Henauer, and A. Rigoni. 2019. Cybersecurity exercises: Wargaming and red teaming. *Next Gen. CERTs* 54 (2019), 44.

[46] Barry Schwartz and Andrew Ward. 2004. Doing better but feeling worse: The paradox of choice. *Positive Psychology in Practice* (2004), 86–104.

[47] Shishir Kumar Shandilya, Saket Upadhyay, Ajit Kumar, and Atulya K. Nagar. 2022. AI-assisted computer network operations testbed for nature-inspired cyber security based adaptive defense simulation and analysis. *Fut. Gen. Comput. Syst.* 127 (2022), 297–308.

[48] Oleg Sheyner, Joshua Haines, Somesh Jha, Richard Lippmann, and Jeannette M. Wing. 2002. Automated generation and analysis of attack graphs. In *IEEE Symposium on Security and Privacy (SP'02)*. IEEE, 273–284.

[49] Maxwell Standen, Martin Lucas, David Bowman, Toby J. Richer, Junae Kim, and Damian Marriott. 2021. CAGE challenge 1. In *IJCAI-21 1st International Workshop on Adaptive Cyber Defense*.

[50] Maxwell Standen, Martin Lucas, David Bowman, Toby J. Richer, Junae Kim, and Damian Marriott. 2021. CybORG: A gym for the development of autonomous cyber agents. arXiv:2108.09118v1

[51] Paul Theron, Alexander Kott, Martin Drašar, Krzysztof Rzadca, Benoît LeBlanc, Mauno Pihelgas, Luigi Mancini, and Agostino Panico. 2018. Towards an active, autonomous and intelligent cyber defense of military systems: The NATO AICA reference architecture. In *International Conference on Military Communications and Information Systems (ICMCIS'18)*. IEEE, 1–9.

[52] Vladislav D. Veksler, Norbou Buchler, Claire G. LaFleur, Michael S. Yu, Christian Lebiere, and Cleotilde Gonzalez. 2020. Cognitive models in cybersecurity: Learning from expert analysts and predicting attacker behavior. *Front. Psychol.* 11 (2020), 1049.

[53] Robert L. West and Christian Lebiere. 2001. Simple games as dynamic, coupled systems: Randomness and other emergent properties. *Cogn. Syst. Res.* 1, 4 (2001), 221–239.

[54] Jeong Do Yoo, Eunji Park, Gyungmin Lee, Myung Kil Ahn, Donghwa Kim, Seongyun Seo, and Huy Kang Kim. 2020. Cyber attack and defense emulation agents. *Appl. Sci.* 10, 6 (2020), 2140.

[55] Jing Zhang, Jun Zhuang, and Victor Richmond R. Jose. 2018. The role of risk preferences in a multi-target defender-attacker resource allocation game. *Reliab. Eng. Syst. Safety* 169 (2018), 95–104.

Received 31 January 2023; revised 26 August 2024; accepted 15 December 2024